INCH-POUND

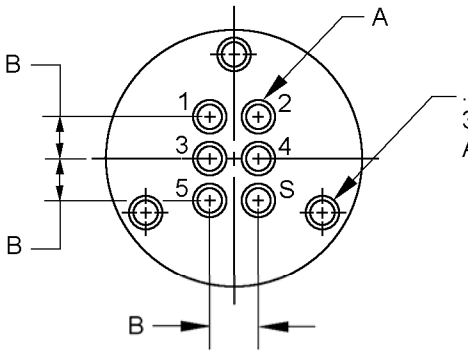
MIL-PRF-27/63D
AMENDMENT 1
12 August 2016
SUPERSEDING
MIL-PRF-27/63D
6 May 2005

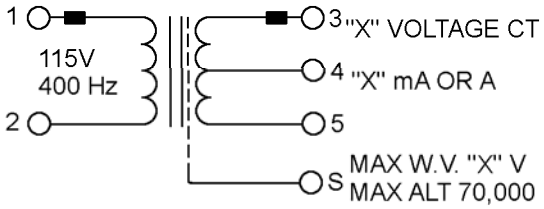
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 50 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

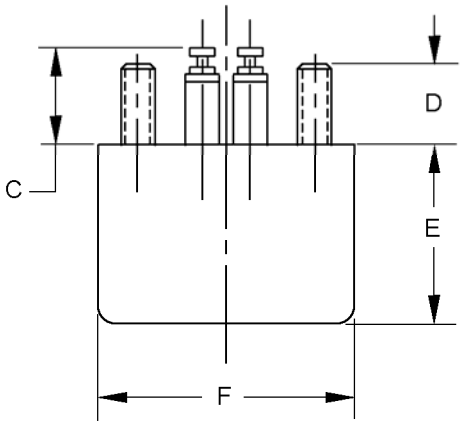
The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.





(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING



Dim	Inches	mm
A	.250 DIA. TYP.	6.35 DIA. TYP.
B	.312	7.92
C	.390 MAX	9.91 MAX
D	.344 ±.031	8.74 ±.078
E	1.500 ±.031	38.10 ±.078
F	2.000 ±.031	50.80 ±.078

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown. Ground terminal "S" is black.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 11 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .390 inch (9.91 mm), maximum.

Terminal strength: MIL-STD-202-211, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 45 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 92 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 3 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 35°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 95°C.

Vibration: MIL-STD-202-204, test condition D.

Marking location: See figure 1.

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VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/63-44 shall be sufficient to grant qualification approval to M27/63-01 through M27/63-44.

Part or Identifying Number (PIN) example:

M	27/63	-01
MIL specification item	Specification sheet no.	Dash no.

TABLE I. Electrical ratings.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01	6	8.333	325
02	6.3	7.937	325
03	7	7.143	325
04	7.5	6.667	325
05	8	6.250	325
06	9	5.556	325
07	10	5.000	325
08	11	4.545	325
09	12	4.167	325
10	12.6	3.968	325
11	13	3.846	325
12	15	3.333	325
13	16	3.125	325
14	18	2.778	325
15	20	2.500	325
16	22	2.273	325
17	24	2.083	325
18	26	1.923	325
19	28	1.786	325
20	30	1.667	325
21	33	1.515	325
22	36	1.389	325
23	39	1.282	325
24	43	1.163	325
25	47	1.064	325
26	51	0.980	325
27	56	0.893	325
28	62	0.806	325
29	68	0.735	325
30	75	0.667	325
31	82	0.610	325
32	91	0.549	325
33	100	0.500	325
34	110	0.455	325
35	115	0.435	325
36	120	0.417	325
37	130	0.385	325
38	150	0.333	325
39	180	0.278	325
40	200	0.250	440
41	220	0.227	440
42	240	0.208	440
43	270	0.185	580
44	300	0.167	580

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204, MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2016-014)

Review activities:
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

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